

Introduction

①

Calculus is the mathematical tool used to analyze changes in physical quantities

OR

Calculus is also mathematics of motion & change

"where there is motion or growth, where variable forces are at work producing acceleration, calculus is right mathematics to apply."

Differential Calculus

"Deals with the problem of finding:

→ Rate of change

→ slope of curve.

Ex: Velocities and acceleration of moving bodies.

* Firing angles that give cannons their maximum range.

* The times when planets would be closest together or farthest apart.

Integral Calculus

Deals with the problem of determining a function from information about its rates of change.

Integral calculus enables us:

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- To calculate length of curves.
- To find areas of irregular region in plane.
- To find the volumes and masses of arbitrary solids.
- To calculate the future location of a body from its present position and knowledge of the forces acting on it.

Real numbers / Number system
5 to 7.

- Natural numbers or positive integers.
- Direction, velocity, acceleration, temperature. therefore we require negative numbers ~~therefore~~ for that negative integers are created, $\dots, -2, -1, 0, 1, 2, \dots$

→ 0

- set of all integers $(\dots, -3, -2, -1, 0, 1, 2, 3, \dots)$.

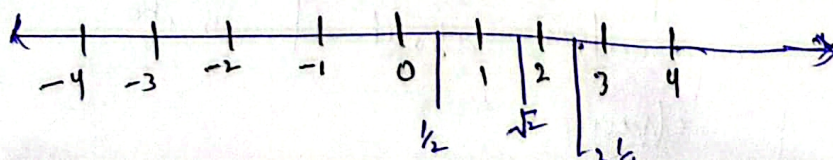
- $\frac{1}{2}$ kg, 0.25 kg. 7th part; 2.5 kg $\Rightarrow$

$$\text{Rational numbers} = \left\{ \frac{p}{q}, q \neq 0; p, q \in \mathbb{N} \right\}.$$

- $\sqrt{2}$, repetition and non-recurring, π is $\frac{22}{7}$ rational approximation.

Every integer is rational number, all +ve and negative integers are rational numbers.

"There is one-to-one correspondence



b/w the set of real numbers and set of points!

Decimal Representation of Real numbers

(3)

"Rational numbers non terminating but repeating."

$$\Rightarrow \frac{4}{3} = 1.333 \dots \quad 3 \text{ repeats.}$$

$$\Rightarrow \frac{4}{3} = 1.\overline{3}$$

$$\Rightarrow \frac{3}{11} = .\overline{27} \quad 27 \text{ repeats.}$$

$$\frac{3}{11} = 0.\overline{27}$$

$$\frac{5}{7} = 0.\overline{714285}$$

$$\frac{22}{7} = 3.\overline{142857}$$

Terminating

$$\Rightarrow \frac{1}{2} = 0.5000 \dots$$

$$\text{or } \frac{1}{2} = 0.5$$

$$\Rightarrow \frac{12}{4} = 3.000 \dots$$

$$\text{or } \frac{12}{4} = 3$$

$$\Rightarrow \frac{8}{25} = 0.32000 \dots$$

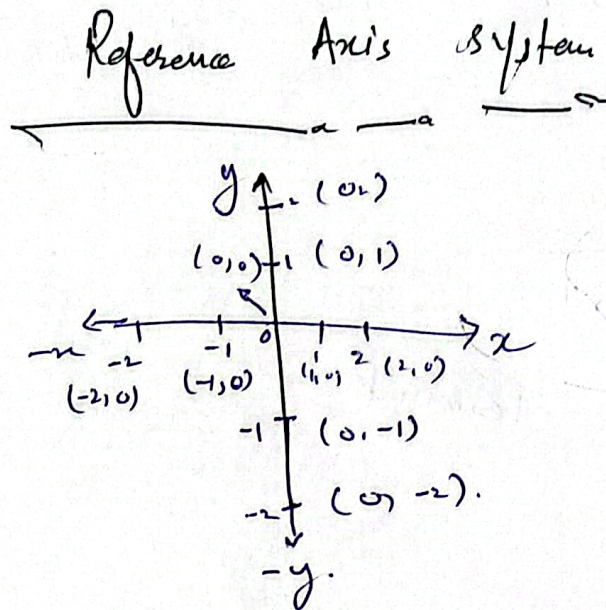
$$\text{or } \frac{8}{25} = 0.32$$

"Irrational Numbers Non Terminating and non repeating".

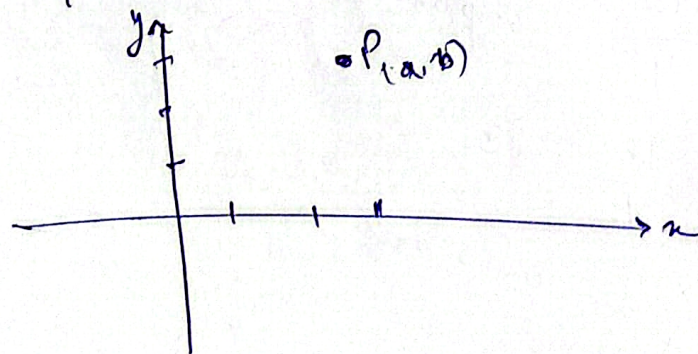
(4)

0.1010010001000010000010000001...

and $\pi = 3.14159265358979323846...$



Coordinate system



3D.

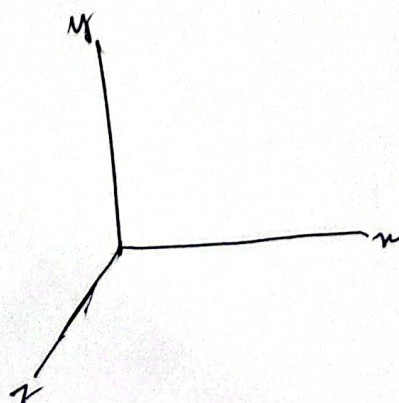
8-octets.

1st (+, +, +)

2nd (-, +, +)

3rd (-, -, +)

4th (+, -, +)



5th (+, +, -)
6th (-, +, -)
7th (-, -, -)
8th (+, -, -)

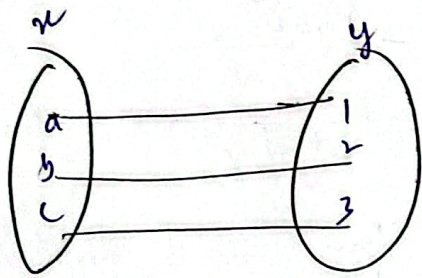
Concept of Function Dependence of one quantity to another (5)

Relation b/w set A and set B - that assigns every element of A with unique element of B.

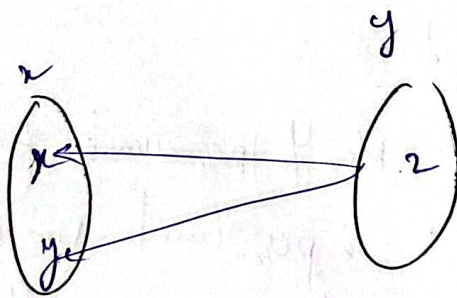
set A is called Domain

set B is called co-Domain

$$y = f(x)$$



30 = computer
30 = student
one to one.
Each student have
each computer.



30 student
15 computer
2 student, for
each computer
onto.

Ex $y = 2x$

$$A = \pi r^2$$

$$V = x^3$$

Profit function

If x is the number of units produced,
cost per unit is Rs 2. The fixed cost is
Rs. 1000 then.

$$C(x) = 2x + 1000 \text{ gives the cost fn.}$$

Example

(6)

if the selling price per unit is Rs. 5

the $R(x) = 5x$

gives the revenue fu.

$$P(x) = R(x) - C(x)$$

$$= 5x - (2x + 1000)$$

$$= 5x - 2x - 1000$$

$$P(x) = 3x - 1000$$

is the profit fu

Piecewise Functions.

Electricity charges are Rs. 4 per unit for the first 100 units and Rs. 5 per unit 100 units to 200 units, Rs. 7 per unit for more than 200 unit

The cost function for x units consumer is given by

$$C(x) = \begin{cases} 4x, & 0 < x \leq 100 \\ 4(100) + 5(x-100), & 100 < x < 200 \\ 4(100) + 5(100) + 7(x-200), & x > 200 \end{cases}$$

Functions of several variables

(7)

Ex: The area of rectangle depends on its length l and width w by the equation.

$$A = lw$$

so we say that

A is a function of l and w

Ex: The volume of a rectangular box depends on the length l , width w , and height h by the equation

$$V = lwh$$

so we say that

V is function of l , w and h .

Ex: The area of a triangle depends on its base length l and height h by the equation

$$A = \frac{1}{2}lh \quad \text{so, we say that}$$

A is a function l and h .

Ex: Volume of right circular cylinder

$$V = \pi r^2 h$$